WORKED SOLUTION

STEP II 2016, Question 13

Probability and Statistics • Normal Approximation

STEP II 2016, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.23, *Stirling from Probability*: https://beyondstep.keithmcnulty.org/ch12/stirling-probability.html

The key idea

A single probability $\mathrm{P}(X = k)$ of a discrete distribution is the area of a bar of width 1 centred on k. If the distribution is close to a normal distribution, that area is roughly the normal density at k times the width 1. Applying this at the mean, where the standard normal density is $\frac{1}{\sqrt{2\pi}}$, turns a known exact probability into an approximation. Choosing the distribution cleverly makes the exact probability contain the factorials we want.

Solution

The approximation we use throughout

Suppose W is integer-valued with mean μ and variance σ^2, and is well approximated by $V \sim \mathrm{N}(\mu, \sigma^2)$. The density of V is

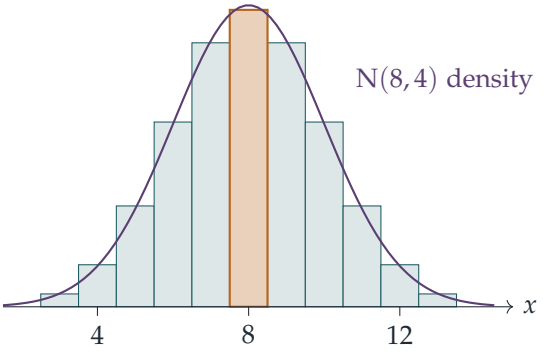
$$f(v) = \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} \, \mathrm{e}^{-\frac{1}{2}\left(\frac{v-\mu}{\sigma}\right)^2}.$$

With a continuity correction,

$$\mathrm{P}(W = k) \approx \mathrm{P}\left(k - \tfrac{1}{2} < V < k + \tfrac{1}{2}\right) = \int_{k-1/2}^{k+1/2} f(v)\,\mathrm{d}v \approx 1 \times f(k),$$

the last step because the interval has width 1 and, when σ is not small, f hardly changes across it. At $k = \mu$ the exponential is $\mathrm{e}^0 = 1$, so

$$\mathrm{P}(W = \mu) \approx \frac{1}{\sigma\sqrt{2\pi}}. \qquad (*)$$



Part (i)

For $X \sim \mathrm{B}(16, \frac{1}{2})$,

$$\mathrm{E}(X) = np = 8, \qquad \mathrm{Var}(X) = np(1-p) = 16 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 4,$$

so $\sigma = 2$. Since $p = \frac{1}{2}$ the distribution is symmetric about 8 and n is reasonably large, so $\mathrm{N}(8, 4)$ is a good approximation (the figure shows the bars against the normal curve). By $(*)$, the shaded bar has area

$$\mathrm{P}(X = 8) \approx \frac{1}{2\sqrt{2\pi}},$$

as required.

Part (ii)

Let $Y \sim \mathrm{B}(2n, \frac{1}{2})$. Then $\mathrm{E}(Y) = n$ and $\mathrm{Var}(Y) = 2n \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{n}{2}$, so $\sigma = \sqrt{n/2}$. Exactly,

$$\mathrm{P}(Y = n) = \binom{2n}{n} \left(\tfrac{1}{2}\right)^n \left(\tfrac{1}{2}\right)^n = \frac{(2n)!}{(n!)^2\, 2^{2n}}.$$

For large n the normal approximation applies, and $(*)$ gives

$$\mathrm{P}(Y = n) \approx \frac{1}{\sqrt{n/2}\,\sqrt{2\pi}} = \frac{1}{\sqrt{n\pi}}.$$

Equating the two expressions and multiplying by $2^{2n}(n!)^2$:

$$(2n)! \approx \frac{2^{2n}(n!)^2}{\sqrt{n\pi}},$$

which is the required result. (Part (i) is the case $n = 8$: $\sqrt{8\pi} = 2\sqrt{2\pi}$.)

Part (iii)

Let $Z \sim \mathrm{Po}(n)$, so $\mathrm{E}(Z) = \mathrm{Var}(Z) = n$ and $\sigma = \sqrt{n}$. Exactly,

$$\mathrm{P}(Z = n) = \frac{\mathrm{e}^{-n} n^n}{n!}.$$

Ask yourself Why may we approximate a Poisson distribution by a normal one here?

A $\mathrm{Po}(n)$ variable has the same distribution as the sum of n independent $\mathrm{Po}(1)$ variables, so by the central limit theorem it is approximately $\mathrm{N}(n, n)$ when n is large. This is the standard rule that $\mathrm{Po}(\lambda) \approx \mathrm{N}(\lambda, \lambda)$ for large λ.

So by $(*)$,

$$\frac{\mathrm{e}^{-n} n^n}{n!} \approx \frac{1}{\sqrt{n}\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi n}},$$

and rearranging,

$$n! \approx \sqrt{2\pi n}\, \mathrm{e}^{-n}\, n^n,$$

as required.

Checks

- Part (i): exactly, $\mathrm{P}(X = 8) = \binom{16}{8}/2^{16} = 12870/65536 \approx 0.1964$, while $\frac{1}{2\sqrt{2\pi}} \approx 0.1995$. That is within about 1.6%.
- Part (ii) with $n = 10$: $\binom{20}{10}/2^{20} \approx 0.1762$ against $1/\sqrt{10\pi} \approx 0.1784$. The ratio of the two sides of the factorial approximation tends to 1 as n grows (about 0.9988 at $n = 100$).
- Part (iii) with $n = 10$: $10! = 3\,628\,800$, and $\sqrt{20\pi}\,\mathrm{e}^{-10}\,10^{10} \approx 3\,598\,696$, under 1% too small.
- Consistency: substituting the result of (iii) for both $(2n)!$ and $n!$ into (ii) gives $\sqrt{4\pi n}\,\mathrm{e}^{-2n}(2n)^{2n}$ on both sides, so the two approximations agree.

Common slips

- Using the standard normal density $\frac{1}{\sqrt{2\pi}}$ without dividing by σ. The density of $\mathrm{N}(\mu, \sigma^2)$ at its mean is $\frac{1}{\sigma\sqrt{2\pi}}$.
- Using the standard deviation $\frac{n}{2}$ instead of $\sqrt{n/2}$ in part (ii), or getting the variance of $\mathrm{B}(2n, \frac{1}{2})$ wrong.
- Forgetting to say why the Poisson distribution may be approximated by a normal one in part (iii).

Beyond STEP

The result of part (iii) is Stirling's formula, and the argument is a heuristic form of the central limit theorem applied at a single point, a "local limit theorem". Making it rigorous needs care: knowing that a distribution function is close to the normal one

does not by itself say that individual probabilities are close to the density. Stirling's formula can also be proved directly, for example by comparing $\ln n! = \sum \ln k$ with $\int \ln x \, \mathrm{d}x$, or from the gamma function by Laplace's method. Blitzstein and Hwang, *Introduction to Probability*, covers the central limit theorem and the Poisson–normal link; Grimmett and Welsh, *Probability: An Introduction*, is a good companion.

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